

PHYSICS-INFORMED BUT NOT PHYSICS-CONSISTENT: ERROR GEOMETRY AND SUBSPACE PROJECTION FOR NEURAL AC POWER FLOW

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ABSTRACT

Recent neural power-flow solvers, including emerging foundation models, achieve accurate voltage predictions, yet such accuracy does not necessarily imply physically consistent solutions. Even small complex voltage errors can yield large AC power-balance residuals. We study this accuracy–consistency gap across PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA on realistic 2224-bus GB-network scenarios, with cross-grid evaluation of GridSFM over 31 systems. Using an SVD basis fitted to training AC power-flow solutions, we find that neural prediction errors contain substantial components outside the dominant solution subspace. To address this mismatch, calibrated solution-subspace projection (CSP) suppresses off-subspace prediction components after train-only bias calibration, reducing Mean PB by 67.0%, 37.8%, 40.5%, and 68.9% for PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA, respectively, relative to calibrated predictions, while improving voltage-magnitude accuracy in all four models. These results identify output-error geometry as an important factor in physics-consistent neural AC power flow.

Index Terms— AC power flow, physics-informed neural network, error geometry, calibration, subspace projection

1. INTRODUCTION

AC power flow (AC-PF) is a fundamental tool for steady-state power-system analysis [1], repeatedly evaluated in time-series and contingency studies [2] and underlying optimal power flow (OPF) [3, 4] and state-estimation workflows [5]. This computational burden has motivated neural surrogates spanning [6, 7] direct data-driven approximation [8, 9], physics- or constraint-aware training [10, 11], learned projection-based correction [12], and feasibility-seeking mechanisms [13, 14]. More recent cross-grid and foundation-model efforts—including GridSFM [15], gridfm-graphkit and GENCO [16, 17], and LUMINA [18]—extend this direction with global context and rich, heterogeneous, topology-aware representations [19] designed to operate across diverse grids and operating conditions [20, 21]. However, accurate voltage prediction does not necessarily imply AC-consistent output [22, 23]. Physics-informed objectives can reduce power-flow residuals [24, 25], but remain soft training objectives and do not by themselves enforce exact satisfaction of the nonlinear AC equations [26]. Consequently, small voltage errors can coexist with non-negligible power-balance violations. This raises a question beyond

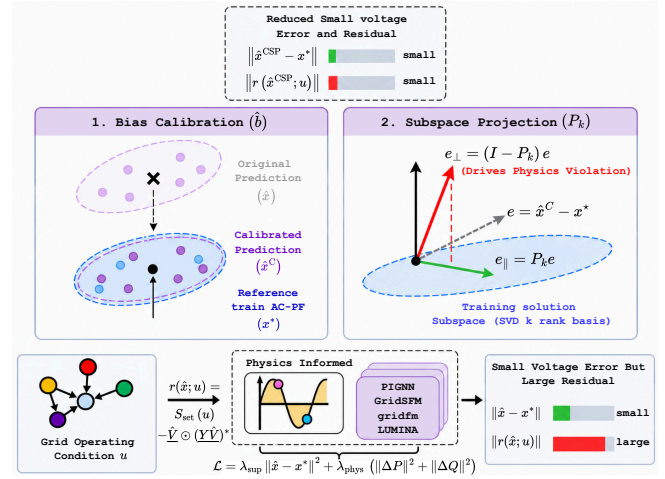


Fig. 1. Bias calibration and solution-subspace projection jointly suppress systematic bias and off-subspace prediction components, improving both voltage accuracy and AC power-balance consistency.

state accuracy: which properties of the remaining voltage error govern AC power-balance residuals?

We hypothesize that the key factor is the *direction* of the remaining voltage error: similarly sized errors can induce very different power-balance residuals depending on their orientation across buses.

Goodwin et al. [27] characterized the differential geometry and linear approximations of the AC-PF solution manifold. We instead study neural-surrogate error geometry empirically, asking whether error orientation relative to a training-derived subspace explains the remaining physical inconsistency. This also differs from Park et al. [28], who use a PCA-based output space during training and to warm-start an exact solver, whereas we use a training-solution subspace to analyze and post-process frozen AC-PF predictions.

We evaluate PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA on realistic 2224-bus GB-network scenarios. From training solutions only, we construct separate SVD subspaces for voltage magnitude and angle and decompose calibrated prediction errors into in- and off-subspace components. Fixed-norm interventions and local Jacobian analysis show that off-subspace errors are systematically more residual-sensitive. Motivated by this observation, we introduce *calibrated solution-subspace projection* (CSP), which combines train-only bias correction with projection onto the dominant training-supported solution directions.

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Across all four models, CSP reduces AC power-balance residuals while improving voltage-magnitude accuracy, with model-dependent effects on angle accuracy. CSP adds negligible computational burden at inference and remains beneficial when used to initialize one Newton refinement step. Additional GridSFM evaluation across 31 systems shows that the benefit extends beyond GBnetwork, whereas distribution-shift experiments reveal that the training-solution subspace can become mismatched to a shifted operating distribution.

Code: github.com/Kimchangheon/neural-acpf-error-geometry

2. ERROR GEOMETRY AND CSP

2.1. AC power-balance consistency and calibration

For an N -bus grid, let $x = [|V|, \theta] \in \mathbb{R}^{2N}$ denote the voltage state, with $\underline{V} = |V| \odot e^{j\theta}$. Given operating condition u , a surrogate predicts $\hat{x} = f_\phi(u)$, from which (P, Q) are recomputed using the network admittance model. For setpoints $(P^{\text{set}}, Q^{\text{set}})$, we define

$$r(\hat{x}; u) = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} P^{\text{set}} - P \\ Q^{\text{set}} - Q \end{bmatrix}, \quad \text{PB} = \sqrt{\Delta P^2 + \Delta Q^2}.$$

PB uses ΔP at PV/PQ and ΔQ at PQ buses, zero elsewhere; prescribed PV/slack magnitudes and the slack angle are restored before scoring. Mean/Max PB aggregate all bus-scenario pairs on a common 100-MVA p.u. base.

Before geometric analysis, we remove train-only prediction bias:

$$\hat{b} = \mathbb{E}_{s \in \text{tr}} [\hat{x}_s - x_s^*], \quad \hat{x}^C = \hat{x} - \hat{b},$$

with $\hat{b}$ frozen at test time; angle offsets use circular differences after gauge alignment.

2.2. Error geometry in a training solution subspace

From training Newton–Raphson solutions, we construct separate rank- k bases for centered magnitudes X_V and gauge-aligned angles X_θ :

$$X_V = W_V \Sigma_V U_V^\top, \quad X_\theta = W_\theta \Sigma_\theta U_\theta^\top, \quad P_k = W_k W_k^\top, \quad (1)$$

where $W_k = \text{diag}(W_{V,k}, W_{\theta,k})$ contains the leading k directions per channel; $k = 16$ denotes 16 per channel. These subspaces capture dominant training-solution variation and define the error coordinates. For calibrated error $e = \hat{x}^C - x^*$,

$$e^\parallel = P_k e, \quad e^\perp = (I - P_k) e, \quad (2)$$

are its in- and off-subspace components. Around a converged state with fixed topology and bus types,

$$r(x^* + e; u) = J(x^*; u) e + \mathcal{O}(\|e\|_2^2), \quad (3)$$

where $r(x^*; u) = 0$ and J is the AC residual Jacobian; hence equal-norm errors can yield different residuals by orientation.

2.3. Calibrated solution-subspace projection

Motivated by this directional sensitivity, CSP first removes train-estimated bias and then projects onto the dominant training-solution directions:

$$\hat{x}^{\text{CSP}} = \bar{x}_{\text{tr}} + P_k \left(\hat{x} - \hat{b} - \bar{x}_{\text{tr}} \right) \quad (4)$$

where $\bar{x}_{\text{tr}}$ is the training-solution mean; both $\bar{x}_{\text{tr}}$ and P_k remain fixed at test time.

Projection can remove prediction error but also discard valid reference variation. Let $e^C = \hat{x}^C - x^*$ and $e^{\text{CSP}} = \hat{x}^{\text{CSP}} - x^*$. Since P_k is orthogonal,

$$\|e^{\text{CSP}}\|_2^2 - \|e^C\|_2^2 = \|(I - P_k)(x^* - \bar{x}_{\text{tr}})\|_2^2 - \|(I - P_k)e^C\|_2^2. \quad (5)$$

Hence, CSP improves state accuracy when the removed off-subspace prediction-error energy exceeds the discarded reference variation.

CSP is a lightweight, data-driven post-hoc output correction, not a projection onto the nonlinear AC-feasible set, and therefore does not guarantee exact feasibility.

3. EXPERIMENTS AND RESULTS

3.1. Experimental setup

Our primary study uses 37,022 converged pandapower Newton–Raphson operating points on the 2224-bus GBnetwork, split 1:1:1 for training/validation/test. The corpus combines global load/generation scaling with independent load P/Q and active-generation perturbations; the backbone topology and PV-voltage control schedule remain fixed. We evaluate PIGNN-GC, our global-context extension of PIGNN-Attn-LS [26], GridSFM [15], gridfm-graphkit [16], and LUMINA [18]; GridSFM is also evaluated cross-grid on 31 systems.

All models are trained from scratch using model-specific stable initialization, and CSP is evaluated within each frozen model. gridfm-graphkit uses its released architecture/configuration without a pretrained checkpoint. GridSFM retains the released backbone but is adapted from AC-OPF to AC-PF by fixing generator dispatch and training only for voltage-state prediction with PF-specific residuals. LUMINA likewise retains its released backbone but uses an enhanced AC-PF configuration with known-variable masking, electrical bus features, and stabilized initialization. All models use pre-solution pandapower [1, 29] internal PPC-derived inputs and a common PPC-based AC power-balance residual evaluator. Training and inference use one NVIDIA H100 96-GB GPU; inference uses the largest fitting batch size. Calibration statistics and solution bases use training data only. From $k \in \{4, 8, 16, 32, 64\}$, validation Mean PB selects $k^* = 16$ without degrading calibrated voltage-magnitude RMSE and is frozen for test and distribution-shift experiments. Unless otherwise stated, mechanistic, ablation, and cross-grid analyses use model seed 42, whereas Tables 1 and 3 aggregate seeds 41–43.

Voltage RMSE alone can be small even when a model largely predicts a static per-bus profile rather than following scenario-dependent voltage variation [15]. To isolate this scenario-dependent response, we center each bus as $\tilde{V}_{is} = V_{is} - \bar{V}_i$ and $\tilde{\hat{V}}_{is} = \hat{V}_{is} - \bar{V}_i$, and report

$$R_w^2 = \text{corr}^2(\tilde{V}_{is}, \tilde{\hat{V}}_{is}), \quad a_w = \frac{\text{Cov}(\tilde{V}_{is}, \tilde{\hat{V}}_{is})}{\text{Var}(\tilde{V}_{is})}. \quad (6)$$

R_w^2 is the squared correlation between the centered prediction and reference pooled over all bus-scenario pairs, while a_w measures the scale of the predicted variation relative to the true variation.

3.2. Off-subspace error and directional sensitivity.

Although the rank-16 basis explains 98.8%/99.7% of training voltage-magnitude/angle variation, the median fractions of total squared error lying outside the respective subspaces are 26.3/55.0/56.8/27.1% for magnitude and 10.3/2.7/39.6/31.7% for angle, for PIGNN-GC/GridSFM/gridfm-graphkit/LUMINA, respectively. Even after accounting for differences in the corresponding total error norms,

Table 1. GBnetwork results at $k = 16$. All models report mean $\pm$ standard deviation over three model seeds. C denotes calibration, P_{16} projection, and CSP_{16} their composition. Inference time is GPU-synchronized forward execution plus the corresponding output transform, reported per scenario.

Model	Output	$ V $ RMSE $\downarrow$	θ RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$	Max PB $\downarrow$	Inference (ms/scen.)
Per-bus mean	Baseline	.02341	6.847	.000	.000	.12084	5.88	–
PIGNN-GC	Raw	.02850 $\pm$.00232	1.108 $\pm$.175	.671 $\pm$.050	.546 $\pm$.072	.31592 $\pm$.03240	195.89 $\pm$ 52.54	10.034 $\pm$.023
	C	.01393 $\pm$.00105	.777 $\pm$.161	.671 $\pm$.050	.546 $\pm$.072	.16661 $\pm$.02352	190.00 $\pm$ 37.53	10.028 $\pm$.023
	P_{16}	.01624 $\pm$.00254	1.091 $\pm$.166	.829 $\pm$.045	.541 $\pm$.071	.07594 $\pm$.00422	5.07 $\pm$.85	10.036 $\pm$.023
	CSP_{16}	.01222 $\pm$.00134	.835 $\pm$.141	.829 $\pm$.045	.541 $\pm$.071	.05497 $\pm$.00257	3.93 $\pm$.07	10.034 $\pm$.032
GridSFM	Raw	.02868 $\pm$.01652	2.294 $\pm$.302	.768 $\pm$.151	.724 $\pm$.149	.26867 $\pm$.13004	9.10 $\pm$ 4.02	25.045 $\pm$.207
	C	.01086 $\pm$.00403	1.781 $\pm$.055	.768 $\pm$.151	.724 $\pm$.149	.08251 $\pm$.01093	7.29 $\pm$ 3.53	25.015 $\pm$.096
	P_{16}	.01814 $\pm$.01145	2.244 $\pm$.325	.936 $\pm$.032	.724 $\pm$.149	.07630 $\pm$.01942	5.08 $\pm$.49	25.087 $\pm$.279
	CSP_{16}	.00782 $\pm$.00307	1.807 $\pm$.053	.936 $\pm$.032	.724 $\pm$.149	.05136 $\pm$.00143	4.94 $\pm$.42	24.577 $\pm$.091
gridfm-graphkit	Raw	.00892 $\pm$.00154	.385 $\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.09282 $\pm$.01763	19.82 $\pm$ 8.65	5.917 $\pm$.019
	C	.00753 $\pm$.00118	.367 $\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.07685 $\pm$.01307	18.88 $\pm$ 8.69	5.730 $\pm$.004
	P_{16}	.00617 $\pm$.00099	.503 $\pm$.018	.959 $\pm$.008	.801 $\pm$.044	.04593 $\pm$.00156	5.36 $\pm$.56	5.723 $\pm$.011
	CSP_{16}	.00606 $\pm$.00090	.496 $\pm$.021	.959 $\pm$.008	.801 $\pm$.044	.04571 $\pm$.00135	5.08 $\pm$.18	5.779 $\pm$.012
LUMINA	Raw	.02653 $\pm$.00031	2.703 $\pm$.313	.443 $\pm$.014	.346 $\pm$.012	1.29896 $\pm$.85056	664.19 $\pm$ 448.00	7.54 $\pm$ 0.37
	C	.01780 $\pm$.00020	1.442 $\pm$.060	.443 $\pm$.014	.346 $\pm$.012	.21432 $\pm$.01186	20.65 $\pm$ 15.82	7.38 $\pm$ 0.16
	P_{16}	.01898 $\pm$.00049	2.160 $\pm$.268	.906 $\pm$.012	.338 $\pm$.012	.07641 $\pm$.00128	5.47 $\pm$.56	7.98 $\pm$ 0.87
	CSP_{16}	.01570 $\pm$.00024	1.282 $\pm$.058	.906 $\pm$.012	.338 $\pm$.012	.06662 $\pm$.00074	4.61 $\pm$.19	7.89 $\pm$ 0.57

scenarios with larger off-subspace fractions tend to have higher Mean PB, with partial Spearman correlations of .659/.635/.594/.833 for magnitude and .789/.906/.809/.860 for angle (all 95% CIs exclude zero). We next test this directional association with an oracle diagnostic using the known reference state, varying the voltage-magnitude error orientation at fixed ℓ_2 norm. We define

$$\tilde{e}(\alpha) = \|e_V\|_2 \frac{e_V^\parallel + \alpha e_V^\perp}{\|e_V^\parallel + \alpha e_V^\perp\|_2}. \quad (7)$$

Decreasing α suppresses off-subspace error; the control analogously suppresses the in-subspace component by exchanging $e_V^\parallel$ and $e_V^\perp$. Phase angles remain at their calibrated predictions, and Mean PB is normalized at $\alpha = 1$. Across all four surrogates, suppressing off-subspace error consistently reduces Relative Mean PB, whereas suppressing in-subspace error produces the opposite trend (Fig. 2). Although the effect size varies across models, its direction is consistent. Because the voltage-magnitude error norm is fixed throughout the intervention, these results show that error orientation affects AC power-balance violation independently of error magnitude.

Consistent with the fixed-norm experiment, a local Jacobian analysis using directional gain $g(d) = \|Jd\|_2/\|d\|_2$ shows that off-subspace perturbations have larger gain than in-subspace perturbations. In a model-independent audit of 4,096 paired random directions at 128 held-out operating points, the median paired off-/in-subspace Jacobian gain ratio is 1,563 $\times$. The same directional sensitivity is present in the actual calibrated prediction errors: the ratios of median off- to in-subspace gains are 22.81 $\times$, 575.83 $\times$, 4.69 $\times$, and 6.51 $\times$ for PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA, respectively. The observed error regime is consistent with the first-order expansion in Eq. (3), with median linearized-to-nonlinear residual-change ratios between 0.996 and 1.009 across all four models.

3.3. Solution subspace specificity.

Matched random, graph-Laplacian, and spatial low-pass projections all degrade voltage-magnitude accuracy and scenario tracking and increase AC residuals relative to the solution-SVD basis (Table 2),

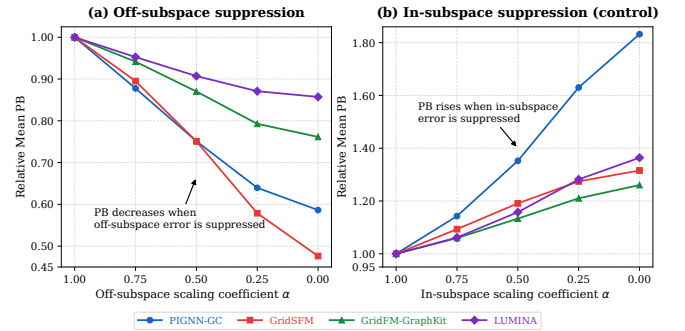


Fig. 2. Fixed-norm directional intervention on GBnetwork, with Relative Mean PB normalized at $\alpha = 1$.

Table 2. Basis controls on GBnetwork for PIGNN-GC ($k = 16$). Random denotes the mean over eight independently sampled bases.

Projection	$ V $ RMSE $\downarrow$	θ RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$
None (C)	.01421	.753	.68522	.49329	.14511
Solution SVD	.01287	.822	.85299	.48903	.05690
Random	.02336	6.825	.00540	.00290	17.74
Laplacian LF	.02132	5.901	.20128	.12304	30.53
Spatial LP	.01568	1.926	.63080	.40717	29.29

showing that CSP gains depend on training-solution geometry rather than generic dimensionality reduction or smoothing.

3.4. Post-training CSP correction

Table 1 compares raw predictions, calibration (C), projection (P_{16}), and CSP_{16} on frozen outputs from PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA. Relative to C , CSP reduces Mean PB by 67.0%, 37.8%, 40.5%, and 68.9%, respectively, while lowering voltage-magnitude RMSE in all four models. Projection improves

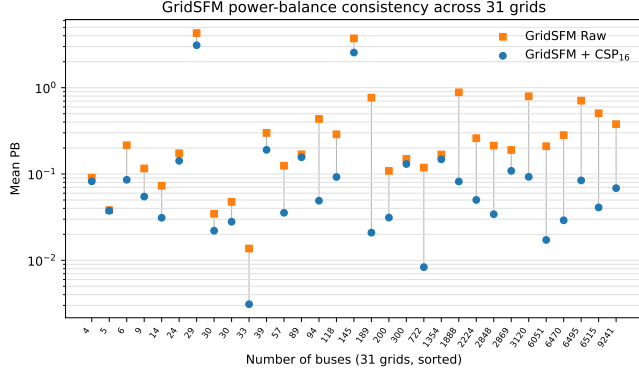


Fig. 3. Grid-by-grid physical consistency of GridSFM before and after CSP₁₆ across 31 systems ordered by bus count. Mean PB is shown on a logarithmic scale.

R_w^2 with little change in a_w ; angle accuracy is model dependent. Raw and CSP16 inference times are comparable.

Magnitude-only, angle-only, and full projection show model-dependent effects of angle projection. PIGNN-GC achieves the lowest PB with full CSP, whereas GridSFM performs best with magnitude-only projection. For gridfm-graphkit, adding angle projection reduces Mean/Max PB from .0612/18.23 to .0457/5.08, but increases angle RMSE from .367° to .496°. For LUMINA, it reduces Mean/Max PB from .1939/38.53 to .0660/4.51 while improving angle RMSE from 1.374° to 1.215°. Thus, better physical consistency from angle projection need not imply better angle accuracy.

These model-dependent accuracy changes are explained by Eq. (5). For magnitude, the removed off-subspace prediction-error energy exceeds the discarded reference variation across all four models. For angle, the balance differs across models: in gridfm-graphkit, angle projection discards more valid reference variation than model error and therefore worsens angle RMSE, whereas the opposite holds for LUMINA.

3.5. Complementarity with one-step Newton refinement.

We test CSP before one Newton update (NR₁) of non-slack angles and PQ voltage magnitudes; validation selects $\eta = 1$ for all models, with no linear-solve failures. CSP initialization further lowers magnitude and angle RMSE in all four models and reduces post-update Mean PB by 21.1–96.4% relative to C initialization (Table 3).

For LUMINA, $C + \text{NR}_1$ exhibits a large one-seed overshoot despite a successful linear solve, whereas CSP₁₆ initialization is consistent across all seeds, yielding Mean PB $1.653 \pm 0.036 \times 10^{-3}$. This further supports CSP as a favorable initializer for Newton refinement.

3.6. Cross-grid breadth and distribution shift

Cross-grid breadth. Fig. 3 evaluates GridSFM with CSP across 31 systems spanning 4–9,241 buses and sub-kV to 750-kV levels: 28 pandapower/PYPOWER-format systems from IEEE, PEGASE, RTE, and GB benchmark families, plus three Common Information Model (CIM)-based networks. PV-voltage setpoints vary for most cross-grid systems, but remain fixed for GBnetwork and grids with conflicting controllers. For each system, calibration and subspace statistics are fitted from its training data, using nominal $k = 16$ per channel and

Table 3. One Newton update from calibrated or CSP₁₆ outputs on GBnetwork. $|V|$ RMSE and Mean PB are reported in 10^{-3} . Values are means $\pm$ standard deviations over three model seeds.

Model	Start	$ V $ RMSE (10^{-3})	θ RMSE (°)	Mean PB (10^{-3})
PIGNN-GC	C	$1.80 \pm .12$	$.119 \pm .013$	$2.84 \pm .46$
	CSP ₁₆	$1.17 \pm .10$	$.073 \pm .009$	$1.48 \pm .13$
GridSFM	C	11.86 ± 6.77	$.681 \pm .380$	2.50 ± 1.54
	CSP ₁₆	$.97 \pm .20$	$.059 \pm .013$	$1.22 \pm .29$
gridfm-graphkit	C	$.943 \pm .232$	$.0843 \pm .0236$	$.941 \pm .325$
	CSP ₁₆	$.496 \pm .085$	$.0277 \pm .0041$	$.742 \pm .080$
LUMINA	C	13.71 ± 9.23	$.671 \pm .038$	46.08 ± 64.15
	CSP ₁₆	$1.040 \pm .016$	$.0802 \pm .0031$	$1.653 \pm .036$

Table 4. PIGNN-GC with fixed-rank CSP₁₆ under topology and operating-distribution shifts. T transfers the control calibration and basis, whereas R refits both on condition-matched training data.

Condition	$ V $ RMSE	$R_w^2 \uparrow$	a_w	Mean	Median	p95	p99	Max
Control	.01281	.8476	.4866	.0581	.0131	.2665	.5462	3.58
$N-1$ (T)	.01314	.8391	.4750	.0603	.0131	.2713	.5632	33.16
$N-2$ (T)	.01288	.8458	.4870	.0617	.0132	.2713	.5721	27.45
GBcorr (T)	.01966	.4819	.4025	.1003	.0231	.4512	.9746	7.00
GBcorr (R)	.01620	.5150	.4247	.0680	.0174	.3031	.6415	4.99

the available SVD components on smaller systems. CSP reduces GridSFM Mean PB across all 31 systems.

Topology and spatial shifts. Under $N-1/N-2$ outages, tracking and bulk PB remain close to control, with larger maxima indicating isolated topology-induced violations and the need for worst-case assessment beyond average error [30]. GBcorr, which applies correlated, spatially varying load-scaling factors across 14 zones while preserving total demand and generation settings, causes a broader shift: transferring both control calibration and basis gives Mean PB ≈ 0.100 p.u. Refitting calibration alone reduces this to 0.0944, basis alone to 0.0740, and both to ≈ 0.068 . The transferred/refitted bases retain 93.55%/98.53% of GBcorr magnitude variation; however, even on converged GBcorr references, they yield Mean PBs of 0.0868/0.0438 p.u., respectively. This reference-only diagnostic isolates subspace mismatch from surrogate error, supporting its role in GBcorr degradation and the benefit of basis refitting.

4. CONCLUSION

This work identifies output-error geometry as an important factor in the accuracy–consistency gap of neural AC-PF surrogates. Fixed-norm and Jacobian analyses reveal strong directional sensitivity of AC residuals, while CSP reduces power-balance residuals and improves voltage-magnitude accuracy on GBnetwork. CSP also improves voltage accuracy and mean power-balance residuals after one Newton update, supporting its complementary role in residual-directed refinement. Cross-grid evaluation shows that the power-balance improvement extends beyond GBnetwork, whereas distribution-shift experiments show that a fixed solution subspace can become mismatched to the shifted operating distribution. These results motivate extending physics-consistent neural surrogates toward broader power-system analysis, estimation, and optimization.

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